EP2108 - Special Theory of Relativity

Assignment 4: Electrodynamics in Covariant form

1. **Maxwell's equations.**

   (a) Show that $\nu = 0$ of $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$ gives Gauss's law, and $\nu = 1$ gives the x-component of the Ampère–Maxwell law.

   (b) Show that the dual tensor $G^{\mu\nu}$ is obtained from $F^{\mu\nu}$ by $\mathbf{E}/c \to \mathbf{B},\ \mathbf{B} \to -\mathbf{E}/c$, and verify that $\partial_\mu G^{\mu 0} = 0$ and $\partial_\mu G^{\mu 1} = 0$ give $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$ and the x-component of Faraday's law.

2. **Deriving the field transformation rules.** Starting from $F'^{\mu\nu} = \Lambda^\mu{}_\lambda \Lambda^\nu{}_\sigma F^{\lambda\sigma}$, derive E'_x, E'_y and B'_z. Then show that for S' moving with velocity $\mathbf{v}$ in an arbitrary direction,

$$\mathbf{E}'_\parallel = \mathbf{E}_\parallel, \quad \mathbf{B}'_\parallel = \mathbf{B}_\parallel, \quad \mathbf{E}'_\perp = \gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})_\perp, \quad \mathbf{B}'_\perp = \gamma\left(\mathbf{B} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}\right)_\perp .$$

3. **Fields of a pure magnetic configuration.** In S there is a uniform field $\mathbf{B} = B_0\,\hat{\mathbf{z}}$ and no electric field. Find $\bar{\mathbf{E}}$ and $\bar{\mathbf{B}}$ in $\bar{S}$ (moving with velocity $v\,\hat{\mathbf{x}}$ relative to S). Verify that $E^2 - c^2B^2$ and $\mathbf{E} \cdot \mathbf{B}$ have the same values in both frames.

4. **Lorentz invariants.** Show that

$$F_{\mu\nu}F^{\mu\nu} = -\frac{2}{c^2}\left(E^2 - c^2B^2\right), \qquad F_{\mu\nu}G^{\mu\nu} = -\frac{4}{c}\,\mathbf{E} \cdot \mathbf{B}.$$

   In frame S, $\mathbf{E} = (0, E_0, 0)$ and $\mathbf{B} = (0, 0, B_0)$, with $E_0 < cB_0$. Find the velocity of a frame $\bar{S}$ in which the electric field vanishes, and the magnitude of $\bar{\mathbf{B}}$ there. What happens when $E_0 > cB_0$?

5. **Field of a uniformly moving point charge.** A point charge q is at rest at the origin of S'. Using the transformation rules (inverted), show that in S (where the charge moves with velocity $v\,\hat{\mathbf{x}}$) the fields are

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q\,\gamma\,\mathbf{R}}{\left[\gamma^2(x - vt)^2 + y^2 + z^2\right]^{3/2}}, \qquad \mathbf{B} = \frac{1}{c^2}\,\mathbf{v} \times \mathbf{E},$$

   where $\mathbf{R} = (x - vt)\hat{\mathbf{x}} + y\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}$. Comment on the shape of the field lines as $v \to c$.

6. **Magnetism from electrostatics (current-carrying wire).** A long straight neutral wire lies along x. In the lab frame S the positive charges (line density $+\lambda$) are at rest, and the negative charges (line density $-\lambda$) move with velocity $+v\,\hat{\mathbf{x}}$. A positive test charge q also moves with velocity $v\,\hat{\mathbf{x}}$ at a distance s from the wire.

(a) Find the force on q in S (purely magnetic).

(b) Go to the rest frame $\bar{S}$ of the test charge. Use the relativistic transformation of charge densities to find the net line charge $\bar{\lambda}_{\text{tot}}$ of the wire in $\bar{S}$, and thence the electric force on q.

(c) Show that the two descriptions agree after transforming the force between frames, and explain how this demonstrates that magnetism is a relativistic effect of electrostatics.

7. Show directly that the quantity

$$c^2\rho^2 - \left(j_x^2 + j_y^2 + j_z^2\right)$$

is invariant under a Lorentz transformation and equals $c^2\rho_0^2$, where ρ_0 is the charge density in the rest frame of the charges.

8. For two inertial frames S and S', with S' moving at speed v in the $+x$ direction relative to S, the charge and current density transformations are

$$\rho = \gamma\left(\rho' + \frac{vj_x'}{c^2}\right), \qquad j_x = \gamma\left(j_x' + v\rho'\right), \qquad j_y = j_y', \qquad j_z = j_z', \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Show that in the nonrelativistic limit $v/c \ll 1$, these reduce to

$$\rho \simeq \rho', \qquad j_x \simeq j_x' + v\rho',$$

and interpret the terms physically.

9. Using the electromagnetic field transformations

$$\begin{aligned} &E_x = E_x', \quad E_y = \gamma(E_y' + vB_z'), \qquad E_z = \gamma(E_z' - vB_y'), \\ &B_x = B_x', \quad B_y = \gamma\left(B_y' - \frac{vE_z'}{c^2}\right), \quad B_z = \gamma\left(B_z' + \frac{vE_y'}{c^2}\right), \end{aligned}$$

find their nonrelativistic limits for $v/c \ll 1$. Explain whether the resulting relations make physical sense.

10. A source charge and a test charge are separated along the direction of the source charge's velocity. Using the Lorentz force and force-transformation laws, show that if the source charge moves at speed v relative to the test charge, toward or away from it, the force on the test charge is

$$F = \left(1 - \frac{v^2}{c^2}\right) F_{\text{C}}, \qquad F_{\text{C}} = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2},$$

where F_{C} is the ordinary Coulomb-force magnitude at separation r. Use the frame in which the source charge is at rest and account for length contraction.

11. A source charge moves at speed v relative to a test charge, with its velocity perpendicular to the line of separation at the instant considered. Using the Lorentz force

and force-transformation laws, show that the force on the test charge at that instant is

$$F = \frac{1}{\sqrt{1 - v^2/c^2}} F_{\mathrm{C}}, \qquad F_{\mathrm{C}} = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}.$$

Compare this result with the force when the source moves along the line joining the charges.

12. Show that, for a given electromagnetic field, there is an inertial frame in which either $\mathbf{E} = 0$ (if $E < cB$) or $\mathbf{B} = 0$ (if $E > cB$) at a given point if and only if $\mathbf{E} \cdot \mathbf{B} = 0$ at that point. That is, if and only if $\mathbf{E}$ and $\mathbf{B}$ are perpendicular, can we find a frame in which either the electric field or the magnetic field vanishes?